

Hadwik Payidiparthi

9640423355 | phadwik65@gmail.com | linkedin.com

EDUCATION

Indian Institute of Technology Kharagpur

B.Tech. (Hons.) in Computer Science and Engineering | CGPA: 9.64 (Till 5th Semester) Aug 2023 – Present

Narayana Junior College

Class XII, TSBIE – 98.6% Hyderabad Jun 2021 – Apr 2023

Viswabharati High School

Class X, SSC – 10 GPA Andhra Pradesh Till Apr 2021

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, HTML/CSS, Bash, AWK, SQL

Databases: PostgreSQL, MySQL

Frameworks & Tools: Flask, pandas, Numpy, Matplotlib, Git, VS Code, Linux, Lex, Yacc, Bison

Core Competencies: Algorithms & Data Structures, Competitive Programming, Object-Oriented Design, Systems Programming

COURSEWORK

Operating Systems, Computer Networks, Database Management Systems, Compilers, Algorithms I & II, Programming and Data Structures, Discrete Structures, Systems Programming, Formal Language and Automata Theory, Computer Organization & Architecture, Probability and Statistics, Linear Algebra, Numerical and Complex Analysis, Switching Circuits and Logic Design, Software Engineering

PROJECTS

Approximate Nearest Neighbour Search | C++, Flask, HTML, CSS

May 2025 – Jun 2025

- Developed a C++ library for efficient K-Nearest Neighbor search, with both brute-force and tree-based methods
- Designed a custom DataVector ADT with dynamic resizing and various vector operations for easy data handling
- Implemented C++ singleton classes for KD-Tree and Random Projection Tree via a TreeIndex base class
- Engineered optimised search structures to accelerate querying on the high-dimensional Fashion-MNIST dataset
- Applied object-oriented design principles to ensure clean, efficient, and reusable code architecture

Text File Compression and Decompression | C++

Jun 2025 – Jul 2025

- Developed a C++ lossless file compression tool using optimal Huffman Prefix codes for high-ratio text compression
- Implemented efficient bit-level I/O, tree construction, and encoding/decoding routines with a CMake build script
- Built a Flask web interface to handle file uploads, run C++ executable via Python's subprocess, return file to user

KBC++: Online Quiz Platform | Under the guidance of Prof. Debasis Samanta

Mar 2025 – Apr 2025

- Built a Flask-based website with a MySQL database and smooth login functionality for users and administrators
- Developed the responsive HTML, CSS, JavaScript frontend to power a dynamic quiz with real-time score tracking
- Implemented multi-browser user sessions and admin dashboard to manage player selection using Flask-MySQLdb
- Integrated MySQL for storage and retrieval of user data and questions, enabling smooth handling of API requests
- Implemented GitHub Actions workflows for unit tests, ensuring consistent code quality and stability before merging

Online Course Management System | Python, Flask, PostgreSQL, HTML5/CSS3/JS

- Built a comprehensive web-based platform managing the complete lifecycle of online education, from course creation to performance analysis.
- Developed distinct functional modules for Students (discovery, enrollment), Instructors (content administration), Admins (moderation), and Analysts.

- Integrated Psycopg2 for backend-database connectivity and designed a normalized relational schema in PostgreSQL.

AWARDS AND ACHIEVEMENTS

- Attained the title of Expert on Codeforces with a peak rating of 1714 (handle: Ace_18).
- Secured an AIR of 164 in JEE ADVANCED 2023, placing in top 0.1% among more than 180,000 candidates.
- Achieved an AIR of 129 in JEE MAINS 2023 with 99.994 percentile among more than 1,700,000 candidates.
- Selected for the KVPY Fellowship (IISc), 2021 by securing an All India Rank of 197 (Stream SA).
- Achieved rank 21 in AP EAMCET 2023, placing in the top 0.01% among 220,000+ candidates.
- Achieved rank 43 in TS EAMCET 2023, placing in the top 0.014% among 320,000+ candidates.
- Ranked in the top 1% in NSEP, NSEC, NSEA and qualified for INPHO, INCHO, INAO in academic year 2022-2023.
- Selected among the Top 60 members (All India) for the Quadeye Excellence Scholarship Program 2025.
- Positioned among the Top 30 performers at my college in NK Securities Research Scholarship Test 2025.
- Regionalist in the IICPC Quantfest 2025.

POSITIONS OF RESPONSIBILITY

Secretary Sports and Games | *Jagadish Chandra Bose Hall of Residence* Aug 2024 – Apr 2025

- Organised a variety of sports events within the hall and promoted participation in inter-hall GC competitions.

Design Team Member | *Spring Fest, IIT Kharagpur* Sep 2023 – Apr 2024

- Collaborated with team members to conceptualize and produce themed posters and props for Spring Fest 2024.
- Supported on-ground event setup and coordination, ensuring smooth execution and timely arrangements.

EXTRACURRICULAR ACTIVITIES

- Participated in Inter-Hall Maths Olympiad and Volleyball General Championship 2025.
- Completed NSO Health & Fitness Program with perfect attendance, reflecting discipline and commitment.
- Participated in Swachhata Hi Seva Campaign, Mental Health Awareness, and Disaster Management workshops 2023-2024.